RR-M-1850
March 8, 1974
SUPERSEDING
Fed. Spec. RR-M-175C
February 2, 1966 and
Fed. Spec. RR-M-190A
May 24, 1966

FEDERAL SPECIFICATION

MEASURE, LIQUID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Liquid measures covered by this specification are of the lipped and swing spout types for dispensing oil and other liquids.

1.2 Classification. Liquid measures shall be furnished in the following types, as specified (see 6.2).

Type I - Spout.
Class A - Rigid.
Class B - Flexible.

Type II - Lipped.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-775 - Steel Sheets, Carbon, Zinc-Coated.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Dcouments, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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3. REQUIREMENTS

3.1 Materials. Material shall conform to the referenced specifications and requirements specified herein. When materials are not definitely specified, they shall be of the quality normally used for the manufacture of high-quality measures.

3.1.1 Steel zinc-coated. Galvanize steel shall conform to type I, class of QQ-S-775.

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3.1.2 Body, top and handle, type I. The body, top, and handle of the type I measures shall be formed from sheet steel not less than 0.0179 inch thick.

3.1.3 Bottom, type I. The bottom of the types I measures shall be made from sheet steel not less than 0.030 inch thick.

3.1.4 Body and bottom, type II. Type II measures shall be fabricated of zinc-coated steel sheet. Thickness and dimensions shall conform to table I.

3.1.5 Spout.

3.1.5.1 Class A. The rigid spout for class A measures shall be seamless tubing or shall have a welded soldered or brazed seam and shall be not less than 0.028 inch thick and 0.821 inch outside diameter.

3.1.5.2 Class B. The flexible spout for class B measure shall be full four wall interlocked, asbestos packed, zinc-coated steel not less than 32 convolution per foot. The end fitting and flexible spout shall be steel not less than 0.800 inch outside diameter.

3.1.6 Valve control mechanism. The valve control mechanism shall consist of steel components.

3.1.7 Washers. Washers shall be formed of material impervious to oil and shall have adequate oil sealing qualities.

3.2 Design. The measures shown are for the convenience of identification and are not intended to preclude the purchase of measures which are otherwise in accordance with the requirements of this specification.

3.2.1 Construction. Measures shall be constructed to hold their rated capacity while standing upright on a level surface. Measures shall be provided with a definite edge, plate, bar, or wire at or near the top of the body to indicate capacity. Capacity shall not vary more than plus or minus 5 percent of the volume of the indicated capacity when tested as specified in 4.4.4. The measures shall not leak when tested as specified in 4.4.2.

3.2.2 Type I.

3.2.2.1 Body. The body shall be fabricated from sheet steel to the dimensions specified in table I. The sides of the body shall be straight or tapered at the option of the manufacturer. The body joint shall be either single or doubled seamed and all seams shall be soldered or brazed. The body seam shall be located at the center line of the handle. The base of the body shall have a rolled edge and shall have a suitable opening in the side, below the bottom to provide clearance for the spout.

3.2.2.2 Bottom. The bottom of the body shall be constructed with a drainage hole to accommodate the spout. The bottom shall be soldered or brazed to the body at a sufficient distance from the base to allow clearance for proper connection and operation of rigid or flexible spouts. The bottom shall be designed to completely drain the measure.

3.2.2.3 Release valve. Type I measures shall be fitted with a release valve assembly for draining the measure. The assembly shall be designed to be operated by a conveniently placed lever or plunger with a bottom head. The release valve shall be capable of being actuated by finger pressure.

3.2.2.4 Lip. Each measure shall be equipped with a curved nonspill or flanged lip, extending completely around the top of the body, at the option of the manufacturer. The lip shall be secured to the body by either a single or double seam and the seams shall be brazed or soldered if the capacity is located in or upon the seam. The outer edge of the lip shall be rolled.

3.2.2.5 Handles. Each measure shall be equipped with a handle of adequate strength to support the fully loaded measure without sagging (see 4.4.1). Upper and lower ends of the handle shall be brazed or soldered to the body. The handle shall provide suitable grip for the hand and thumb and be without sharp edges that would be injurious to the hand when carrying full capacity of measure. The same size handle shall be used for the 1-, 2-, 4-, or 8-quart capacities.

3.2.2.6 Spouts.

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3.2.2.6.1 Class A. The spout shall be rigid and constructed with a gradual 90° bend and self-adjusting, leak proof swivel for attachment to the bottom of the body, on one end. The other end of the spout shall have a bend of 10° minimum to facilitate pouring. The spout shall be designed to swing in an arch of 160° minimum (see 4.4.3). Length of the spouts shall be as specified in table I.

3.2.2.6.2 Class B. The spout shall be flexible and suitable for reaching locations that would otherwise be inaccessible. One end of the spout shall be secured to the bottom of the body by brazing or soldering. The other end shall be fitted with a curved section of rigid steel tubing to facilitate pouring. The rigid end section shall be equipped with a fastening device so the spout may be attached in a vertical position to the lip of the body. The spout shall be capable of flexing a minimum of 60° without straining the consolutions when tested in accordance with 4.4.3. Length of the spout shall meet the requirements in table I.

TABLE I. Dimensions, type I, class A and B

Capacity of measure	Tolerance in capacity		Body dimensions, minimum		Length of spout minimum	
	Plus	Minus	Diameter of bottom	Overall height	Class A	Class B
Quarter	Drams	Drams	Inches	Inches	Inches	Inches
1	2	1.0	4-3/8 to 4-3/8	6-1/2	13-1/2	10
2	3	1.5	4-3/4 to 5-1/4	8 to 10-1/4	13-1/2	14-3/4 to 15
4	4	2	6 to 6-5/8	9-3/8 to 12-1/2	13-1/2	16 to 17
8	6	3	8-1/8	14-1/8	13-1/2	16-1/2

3.2.3 Type II.

3.2.3.1 Body and bottom. Body and bottom shall be fabricated by either one piece seamless construction or two piece construction with double seam joints. Single seam joints may be used in place of double seam joints in the 2-quart size measures. The body seam shall be located at the centerline of the handle. Seams shall be not less than 1/8 inch wide (measured from the offset) and shall be soldered.

3.2.3.2 Handle. The handle for the measure shall be fabricated of the same material as specified for the body and shall be not less than the metal thickness of the body (see table II). Both edges of the handle shall be beaded over galvanized steel wire. Upper and lower ends of the handle shall be riveted or soldered to the body. When tested as specified in 4.4.1 the handle shall not be deformed or separated from the body.

3.2.3.3 Lip. Lip shall extend completely around the top of the body in accordance with the dimensions specified and fabricated of cold rolled steel sheet of the same thickness of the body specified in table II. The lip shall be secured to the body by either single or double seam then soldered.

TABLE II. Dimensions, type II

Capacity of measure	Minimum thickness of metal	Body dimensions minimums	
	Steel	Diameter at bottom	Over all height
2-quart	.014	4-3/4 to 5-3/8	6-7.8 to 8-1/8
4-quart	.014	5-3/4 to 6-5/8	9 to 10-1/8

3.2.4 Soldering. All soldered joints shall be complete, sound, smooth, clean and shall be free from pin holes and flux residue.

3.3 Marking. The capacity of the measures shall be conspiuously, legibly and permanently indicated on the side of the measure. All measures shall be clearly marked with the manufacturer's name or trademark.

3.4 Finish. All liquid measures shall be finished with a coat of oil resistant enamel, or coated with metal powder (aluminum, brass, or bronze) in an enamel oil base and baked at the manufacturer's option. All metal, regardless of type of treatment, shall be free of foreign matter which might adversely affect coating adhesion.

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3.5 Workmanship. All enameled surfaces of the finished measures shall be free from uncoated areas. The finished measures shall be free from projections, sharp edges, burrs, slivers, or defects affecting appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and service conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 End item inspection.

4.2.1.1 Visual examination. Examination of the end item shall be in accordance with the classification of defects listed in table III. A lot shall consist of all liquid measures of the same type, and size offered for inspection at the same time.

4.2.1.2 Dimensional examination. Examination shall be made of the measure for defects in dimensions. Any dimension not as specified shall be classified as a defect.

4.2.1.3 Inspection levels and AQL's. The acceptable quality levels (AQL's), expressed in defects per 100 units, and inspection levels shall be as follows:

Examination	Inspection level	AQL	
		Major	Minor
Visual	II	2.5	4.0
Dimensional	S-2	4.0	---
End Item Testing	S-2	4.0	---

TABLE III. Classification of defects

Defect	Major	Minor
Dimensions not in accordance with table.....	X	
Type and capacity not as specified.....	X	
Area of corrosion.....		X
Crack, hole, or crevice, sharp edges and projections....	X	
Indicator for capacity omitted, or not at or near top of body.....		X
Body and bottom construction more than 2 pieces.....	X	
Uncoated areas (bare metal shows).....	X	
Capacity marking omitted, not conspicous, not permanent, not located as specified.....	X	
Lip to body seam not soldered.....	X	
Soldering or brazing missing, not adherent, complete, sound, and clean.....		X
Damage or defects affecting function or serviceability..	X	
Damage or defects not affecting function or serviceability.....		X

4.3 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table IV. For examination interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operation. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

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TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling distortion of container.
Weight	Weight of container exceeds the specified limitation.

4.4 Tests.

4.4.1 Handle. A substance shall be placed or poured into the measure equal in weight to one and one-half that of the quantity of water required to fill the measure. (Weight of water is 2.086 pounds per quart.) The measure shall then be lifted by the handle in a level position and suspended in mid-air for a period of a minute. The measure shall be considered as not meeting the requirements of this specification if there is a permanent set or break in the handle (see 3.2.2.5 and 3.2.3.2).

4.4.2 Leakage test. The measure shall be filled to its capacity with water or grade 10W oil for not less than 5 minutes for compliance with 3.

4.4.3 Swivel and flexibility of spouts. Class A spouts shall be measured by any suitable method to determine swivel of 160° and class B spouts shall be measured for the 60° angle bend of flexibility in flexed position (see 3.2.2.6.1 and 3.2.2.6.2).

4.4.4 Capacity test. The capacity of the measure shall be tested by measuring the volume of water necessary to maintain the level indicated. The results shall then be examined to determine compliance with 3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Measures of like description, in quantities as specified (see 6.2) shall be packaged in a close-fitting for conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.1.2 Level C. Measures shall be packaged in accordance with the manufacturers standard practice providing that this insures protection for the measures during shipment and safe delivery to their destination.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Measures of like description, packaged as specified in 5.1 shall be packed in quantities as specified (see 6.2) in a close-fitting box conforming to PPP-B-636, class weather-resistant. Close, seal and strap the boxes in accordance with the appendix to PPP-B-636. The gross weight of the box shall not exceed the weight limitations of the box specification for the grade selected.

5.2.2 Level B. Measures of like description packaged as specified in 5.1 shall be packed in quantities as specified (see 6.2) in a close-fitting box conforming to PPP-B-636, class domestic. Closure shall be in accordance with method II of the appendix to PPP-B-636. The gross weight shall not exceed the weight limitation of the box specification.

5.2.3 Level C. Measures of like description, packaged as specified in 5.1 shall be packed in containers that will assure safe arrival at distination in compliance with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

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5.3 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Measures are intended to be used for measuring and pouring oil and other liquids.

6.2 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of measure required (see 1.2).
(c) Size required (see table I and II).
(d) Coating of plating if different from 3.5.
(e) Selection applicable levels of packaging and packing required (see 5.1 and 5.2).

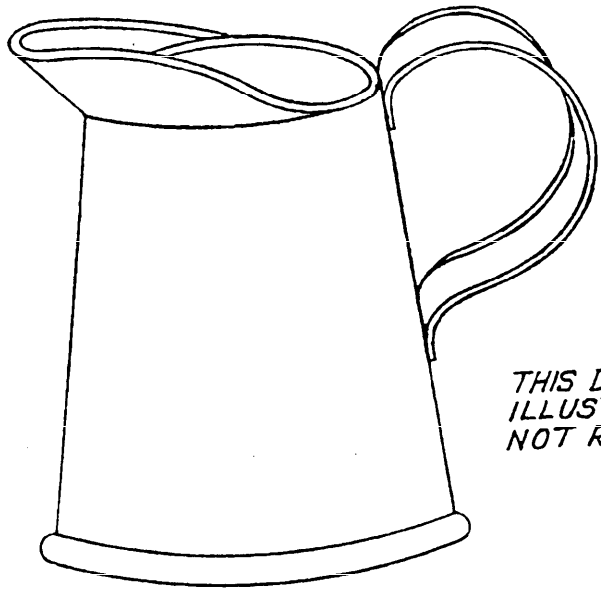
6.3 Supersession data. The specification supersedes the following specifications in the their entirety:

RR-M-175C RR-M-190A

TABLE V. Cross reference (RR-M-175C and RR-M-190A to RR-M-1850)

RR-M-175C		RR-M-190A		RR-M-1850		
Class	Size	Type	Quarts	Type	Class	Capacity
A	1/2 pt. 1 pt. 1 quart 2 quarts 4 quarts	I	1 2 3 4	I	A	1 2 3 4
B	1/2 pt. 1 pt. 1 quart 2 quarts 4 quarts	II	1 2 3 4	I	B	1 2 3 4
				II		2 4

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FIGURE 1. MEASURE, LIQUID, LIPPED.

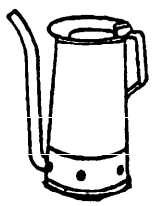


FIGURE 2. Measure, liquid, rigid spout

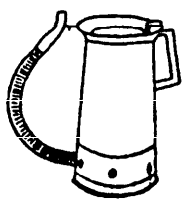


FIGURE 3. Measure, liquid, flexible spout

Preparing activity:

GSA-FSS

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